IMU_CARRIER fab notes

NOT RELEASED: model/CI only; bench gates OPEN.
Board status: SB2_ROUTED_MECH_FIT_OPEN_NOT_FOR_FABRICATION. Every value below is a CI/model output of pcb/fab_export.py and pcb/board_gen.py; no hardware or bench gate is closed by it.

Board

Outline (Edge.Cuts of IMU_CARRIER.kicad_pcb): 40.280 x 14.500 mm, area 427.9 mm2; lower-left corner at X 100.000, Y -114.500 mm in the Gerber/CPL frame (KiCad absolute origin, Y up).
Holes: NPTH 3.20 mm (2 mounting holes; NPTH unplated and without net).

Fabrication

Copper layers: 2; finished thickness 1.6 mm (JLC tolerance +/-10 % for boards >= 1.0 mm).
Stack-up: JLC standard 2-layer FR-4, outer copper 1 oz (https://jlcpcb.com/capabilities/pcb-capabilities, read 2026-11-07).
Impedance: no RF50 net on this board; order without impedance control.
Surface finish, solder mask and legend colours: order options, not fixed by this fab set.
KiCad 10.0.7 DRC on the committed board: 0 errors, 0 unconnected, 20 reviewed warnings, ignored checks [] (report.json; CI/model result).

Files (fab/)

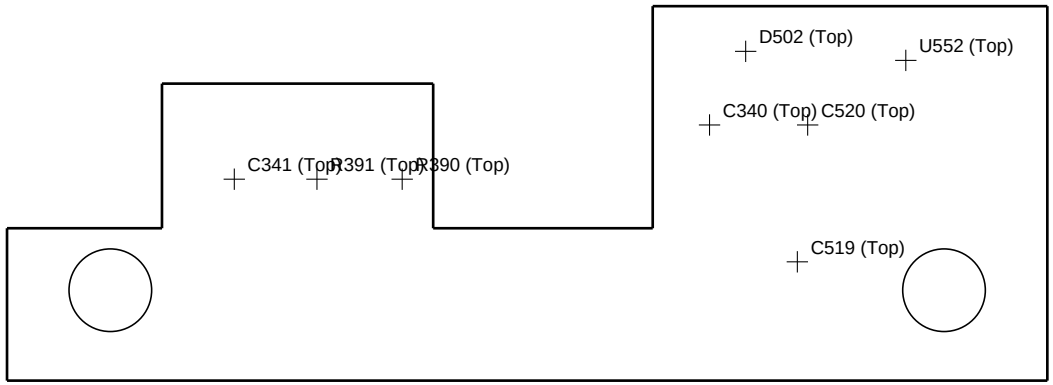
gerber/: Gerber X2 with netlist attributes, Protel extensions, one file per layer, Gerber job file (.gbrjob). drill/: Excellon PTH and NPTH files (mm, decimal, absolute origin, alternate oval mode) and Gerber X2 drill maps. IMU_CARRIER.ipc356: IPC-D-356 netlist of every net with a pad. assembly/: JLC BOM and CPL, the raw KiCad position file, the rotation offset table and the instance map.

Assembly

Boards to build: 1 = instances (IMUPACK 1) + spares 0; every board is the same design and is assembled the same.
Sides with parts: Top (SMD 8).
CPL: Mid X / Mid Y are the KiCad footprint origins (JLC defines them as the component centroid) and Rotation the raw KiCad rotation (degrees, counter-clockwise as the JLC CPL column). JLC rotation and centroid offsets per footprint are TBD (assembly/IMU_CARRIER-cpl-rotation-offsets.csv): check every part in the JLC viewer at order time; nothing here verifies them.

Not proved here (OPEN)

- that the schematic is right
- current capacity or temperature rise beyond the chosen class widths
- impedance, RF loss or isolation, EMC or signal integrity
- creepage beyond the encoded clearances
- that the enclosure fits (only outline and hole checks; heights are not



Board extent 40.280 mm (X) x 14.500 mm (Y)

Top view, Edge.Cuts and holes of IMU_CARRIER.kicad_pcb; crosses = CPL positions (KiCad footprint origins). Not to a fixed scale.
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checked here)
- that the fab's DFM accepts the board, or that CPL rotations are right
- that the land patterns match the purchased MPNs
- any function: hardware and bench gates stay OPEN
Sources (read 2026-11-07): https://jlcpcb.com/help/article/bill-of-materials-for-pcb-assembly; https://jlcpcb.com/help/article/pick-place-file-for-pcb-assembly; https://jlcpcb.com/help/article/how-to-generate-the-bom-and-centroid-file-from-kicad; https://jlcpcb.com/help/article/how-to-generate-gerber-and-drill-files-in-kicad-8; https://jlcpcb.com/capabilities/pcb-capabilities